

BITSAT 2026 — Official Syllabus

Birla Institute of Technology & Science Admission Test | B.E. / B.Pharm | Computer Based Test

■ Exam Pattern

Part	Subject	Questions	Marks
Part I	Physics	30	90
Part II	Chemistry	30	90
Part III	English Proficiency	10	30
	Logical Reasoning	20	60
Part IV	Mathematics / Biology	40	120
Total		130	390

Marking: +3 correct / -1 wrong / 0 unanswered. Duration: 3 hours (no section time limit). Language: English only.

Bonus: Answer all 130 → get 12 extra questions (3 each from Physics, Chemistry, Math/Bio, Logical Reasoning). Cannot revise original answers after starting bonus.

■ Eligibility Criteria

Criterion	Requirement
Year of Exam	Appearing in Class 12 in 2026 OR passed in 2025. Students who passed in 2024 or earlier are NOT eligible.
Board	Recognized Central or State Board under 10+2 system (or equivalent)
Subjects Required	Physics, Chemistry, Mathematics/Biology, one Language, + any additional subject (minimum 5)
Minimum Marks	75% aggregate in PCM/PCB; at least 60% individually in Physics, Chemistry, and Mathematics/Biology
English	Sufficient proficiency in English required
PCB Students	Can apply only for B.Pharm and B.E. Environmental & Sustainability Engineering
Note	Students currently studying at any BITS campus are NOT eligible

■ PHYSICS — 18 Units · 30 Questions · 90 Marks

1. Units & Measurement	SI units, Dimensional analysis, Significant figures, Vernier callipers, Screw gauge, Physical balance
2. Kinematics	Vectors, Position/velocity/acceleration, Constant acceleration, Projectile motion, Circular motion, Relative motion
3. Newton's Laws of Motion	Free body diagrams, Inclined plane, Pulley systems, Centripetal force, Inertial & non-inertial frames
4. Impulse & Momentum	Conservation of momentum, Collisions (elastic/inelastic), System of particles, Centre of mass
5. Work & Energy	Work-Energy theorem, Power, Conservative forces, Potential energy, Conservation of mechanical energy
6. Rotational Motion	Angular kinematics, Moment of inertia, Parallel/perpendicular axes theorems, Torque, Angular momentum, Rolling
7. Gravitation	Newton's law, Gravitational PE, Escape velocity, Kepler's laws, Satellite motion
8. Mechanics of Solids & Fluids	Elasticity, Pressure, Archimedes' principle, Viscosity, Surface tension, Bernoulli's theorem
9. Oscillations	SHM kinematics, Spring-mass system, Pendulums, Forced & damped oscillations, Resonance
10. Waves	Progressive waves, Standing waves, Superposition, Beats, Doppler effect
11. Heat & Thermodynamics	Kinetic theory, Specific heat, Heat transfer (conduction/convection/radiation), First & second laws, Carnot engine
12. Electrostatics	Coulomb's law, Electric field, Potential, Gauss's law, Dipole, Capacitors with & without dielectrics
13. Current Electricity	Ohm's law, Joule heating, Kirchhoff's laws, Potentiometer, Wheatstone bridge, Resistivity & temperature dependence

14. Magnetic Effect of Current	Biot-Savart's law, Ampere's law, Lorentz force, Current loop, Galvanometer conversion
15. Electromagnetic Induction	Faraday's law, Lenz's law, Eddy currents, Self/Mutual inductance, Transformers, AC circuits, LCR
16. Optics	Reflection, Refraction, Lenses, Mirrors, Telescope & Microscope, Young's double slit, Thin film interference, Diffraction, EM spectrum, Polarization
17. Modern Physics	Photoelectric effect, de Broglie wavelength, Rutherford's experiment, Bohr's model, Hydrogen spectrum, Radioactivity, Nuclear fission & fusion
18. Electronic Devices	Energy bands, P-N junction diode, LED, Zener diode, Transistor as amplifier/switch, Logic gates (OR/AND/NOT/NAND/NOR)

■ CHEMISTRY — 14 Units · 30 Questions · 90 Marks

1. States of Matter	SI units, Mole concept, Gas laws, Ideal gas equation, van der Waals equation, Liquid state, Crystal systems, Solid state, Band theory
2. Atomic Structure	Subatomic particles, Bohr model, Quantum mechanics, de Broglie, Uncertainty principle, Orbitals (s/p/d), Aufbau, Hund's rule, Pauli's principle
3. Periodicity & Bonding	Periodic table, s/p/d/f blocks, Periodic trends, Ionic & covalent bonds, VSEPR, Hybridization, MOT, Dipole moment, Hydrogen bond
4. Thermodynamics	First law (enthalpy, Hess's law), Second law (Gibbs energy, spontaneity), Third law introduction
5. Physical & Chemical Equilibria	Raoult's law, Colligative properties, van't Hoff factor, KP & KC, Le Chatelier's principle, pH, Buffer, Henderson equation, Ksp
6. Electrochemistry	Redox reactions, Standard electrode potentials, Nernst equation, Galvanic & electrolytic cells, Faraday's laws, Kohlrausch's law, Corrosion
7. Chemical Kinetics & Surface Chemistry	Rate expressions, Half-life, Arrhenius equation, Collision theory, Adsorption, Catalysis, Colloids, Tyndall effect
8. Hydrogen & s-Block Elements	Hydrogen (hydrides, H ₂ O ₂), Alkali metals (Na, K compounds), Alkaline earth metals (Ca, Mg compounds)
9. p, d & f-Block Elements	Groups 13-18 (properties, key compounds), d-block (transition metals, K ₂ Cr ₂ O ₇ , KMnO ₄ , Fe/Cu/Zn extraction), f-block (Lanthanoids, Actinoids)
9.9 Coordination Compounds	Ligands, IUPAC nomenclature, Werner's theory, VBT, CFT, Isomerism (including stereo), Biological systems (chlorophyll, Vit B ₁₂ , haemoglobin)
10. Organic Chemistry & Hydrocarbons	Functional groups, IUPAC, Inductive/Resonance/Hyperconjugation, Alkanes, Alkenes, Alkynes, Aromatic hydrocarbons, Haloalkanes
11. Stereochemistry	Ethane conformations (Newman & Sawhorse), Geometrical isomerism in alkenes
12. Organic Compounds (O & N)	Alcohols, Phenols, Ethers, Aldehydes, Ketones, Carboxylic acids, Amines, Diazonium salts, Cyanides — nomenclature, preparation, reactions
13. Biological, Industrial & Environmental Chemistry	Carbohydrates, Proteins, Nucleic acids, Vitamins, Polymers (rubber, nylon, polyesters, Bakelite), Pollution, Chemicals in medicine & food
14. Experimental Chemistry	Volumetric analysis, Qualitative salt analysis, Physical chemistry experiments, Purification methods, Organic compound analysis

■ ENGLISH PROFICIENCY & LOGICAL REASONING — 30 Questions · 90 Marks

English: Grammar	Determiners, Prepositions, Modals, Adjectives, Relative pronouns, Tense, Voice, Transformation, Parallel construction
English: Vocabulary	Odd word, Spelling, Synonyms, Antonyms, Phrasal verbs, Idioms, Analogy, Contextual meanings
English: Reading Comprehension	Main ideas, Conclusions from evidence, Facts vs Opinions, Figurative language, Summary, Referents, Jigsaw reading
English: Composition	Word order, Paragraph unity, Topic sentences, Linkers & connectives
Verbal Reasoning	Analogy, Classification (odd one out), Series completion, Logical deduction from passages, Chart logic
Non-Verbal Reasoning	Pattern perception, Figure formation & analysis, Paper cutting, Figure matrix, Rule detection

■ MATHEMATICS — 12 Units · 40 Questions · 120 Marks

1. Algebra	Complex numbers, Quadratic equations, AP/GP, Logarithms, Permutations & Combinations, Binomial theorem, Matrices & Determinants, Sets/Relations/Functions, Mathematical induction, Linear inequalities
2. Trigonometry	Trigonometric functions & graphs, Solution of trig equations, Inverse trig functions
3. 2D Coordinate Geometry	Straight lines, Circles, Conics (Parabola, Ellipse, Hyperbola) — eccentricity, foci, directrices
4. 3D Coordinate Geometry	Direction cosines, Lines in space, Skew lines, Equation of plane, Angle between lines/planes
5. Differential Calculus	Limits, Continuity, Derivatives (all function types), Chain rule, Tangents & normals, Maxima/Minima, Rolle's theorem, MVT
6. Integral Calculus	Indefinite & definite integrals, Integration by substitution/parts/partial fractions, Area under curves
7. Ordinary Differential Equations	Order & degree, Variables separable, Homogeneous DE, Linear first order DE
8. Probability	Addition & multiplication rules, Conditional probability, Bayes' theorem, Discrete random variables, Distributions
9. Vectors	Direction ratios/cosines, Dot & cross products, Scalar triple product
10. Statistics	Measures of dispersion, Frequency distribution analysis
11. Linear Programming	Terminology, Graphical method, Feasible/infeasible regions, Optimal solutions
12. Mathematical Modelling	Real life problems solved using matrices, calculus & linear programming

■ BIOLOGY — 10 Units · 40 Questions (PCB) · 120 Marks

1. Diversity in Living World	Taxonomic categories, Two/Five-kingdom systems, Plant kingdom (Algae to Angiosperms), Animal kingdom (Non-chordate to Class level)
2. Cell: Structure & Function	Cell organelles (ER, Golgi, Mitochondria, Plastids, etc.), Prokaryotic vs Eukaryotic, Mitosis & Meiosis, Biomolecules, Enzymes
3. Genetics & Evolution	Mendelian inheritance, Chromosomal theory, Incomplete/Co-dominance, Linkage & crossing over, DNA replication, Gene expression, Modern Darwinism
4. Structure & Function — Plants	Morphology, Plant tissues, Absorption & movement of water, Mineral nutrition, Photosynthesis (C3/C4), Respiration (TCA cycle, ETS)
5. Structure & Function — Animals	Digestion & respiration, Body fluids & circulation (double circulation), Excretion (urine formation), Locomotion, Nervous & endocrine systems
6. Reproduction, Growth & Movement in Plants	Asexual/sexual reproduction, Gametophytes, Pollination, Fertilization, Embryo development, Growth regulators, Photoperiodism, Vernalisation
7. Reproduction & Development in Humans	Male/Female reproductive systems, Menstrual cycle, Fertilization, Embryo development, Pregnancy, Birth control
8. Ecology & Environment	Ecological levels, Ecosystems, Energy flow, Food chains/webs, Biodiversity, Pollution, Greenhouse effect, Ozone depletion, Deforestation
9. Biology & Human Welfare	Animal husbandry, Communicable diseases (fungi/bacteria/virus/protozoa), Cancer, AIDS, Adolescence, Immunology, Plant breeding, Tissue culture
10. Biotechnology & Applications	Recombinant DNA technology, Microbial technology, Gene therapy, Insulin/Vaccines/Growth hormone production, GM crops, Bt crops